Direction-Aware Offset Enhancement and Dynamic Deformable Convolution for Infrared Small-Target Detection

Baofeng Ji, Dewei Wang, Jianpeng Bai, Guoqiang Zheng, Tongshuai Xin, and Huitao Fan

***Abstract*—Infrared small-target detection (IRSTD) plays a crucial role in both military and civilian applications. However, infrared small targets themselves are extremely tiny in size, often smaller than the convolutional kernel size. Traditional convolution operations tend to introduce background noise, leading to suboptimal performance. In this article, we propose a network named the enhanced dynamic deformable network for IRSTD, which reduces the introduction of background noise during feature extraction while fully extracting small-target features to improve detection performance. Specifically, we design a dynamic deformable convolution to precisely extract features of infrared small targets while minimizing interference from background noise. To enhance the offset capability of convolution kernel sampling points, we propose a direction-aware offset enhancement (DAOE) module, which strengthens the sampling points' ability to bias toward small targets. Furthermore, we introduce a semantic refinement (SR) module at the encoder–decoder transition to enhance deep semantic features, ensuring the network retains semantic awareness while focusing on small targets. Extensive experiments are conducted on three widely used public datasets: NUAA-SIRST, NUDT-SIRST, and IRSTD-1K. The results demonstrate that our proposed method outperforms existing state-of-the-art (SOTA) approaches and can extract more effective target information under complex backgrounds. Our code will be made public at: https://github.com/wdw0537/EDDNet**

***Index Terms*—Deep learning, deformable convolution, feature fusion, infrared small-target detection (IRSTD).**

I. Introduction

INFRARED thermal imaging (ITI) captures target images by detecting thermal radiation through infrared detectors.

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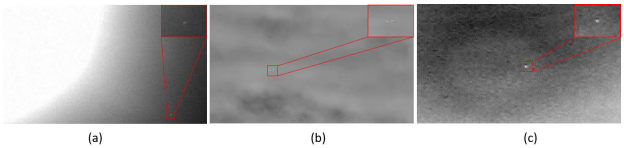


Fig. 1. Sample images from the NUAA-SIRST dataset are presented. (a) Small target measures 6 pixels in size. (b) Contains small targets measuring 6 and 8 pixels, respectively. (c) Small target of 6-pixel dimension. Conventional convolution operations inevitably introduce background noise in these scenarios.

Infrared radiation, with wavelengths surpassing visible light, effectively penetrates atmospheric obstructions like fog and smoke, demonstrating superior detection performance under adverse weather conditions. Thermal signatures derived from infrared sensing enable object identification beyond visual limitations.

The ITI system executes three operational phases: thermal sensor-based radiation acquisition, signal conversion to electrical data, and temperature distribution mapping through thermal image generation. Remote sensing applications yield background-dominated infrared images where distant targets (aircraft, vessels, vehicles) occupy minimal pixel areas, as illustrated in Fig. 1.

Infrared small-target detection (IRSTD) identifies low-signature targets in complex backgrounds, particularly in drone-mounted long-range ITI systems. IRSTD applications span civilian domains including intelligent traffic management [1], agricultural surveillance [2], and subaquatic reconnaissance [3]. Its all-weather functionality further enables critical military deployment in combat engagement and defense systems [4]. Compared with visible-light detection, infrared-based recognition offers enhanced scientific potential in spectral utilization.

Infrared small targets exhibit three distinctive characteristics compared to conventional detection objects.

1) *Low Conspicuity:* Background noise and clutter in infrared images frequently obscure targets, leading to difficulty in discerning authentic targets.
2) *Minimal Pixel Occupancy:* Typically spanning 1–10 pixels due to extended imaging distances between camera and targets.
3) *Shape Variability:* Target morphology and dimensions fluctuate across operational scenarios, influenced by target type and environmental conditions.

IRSTD has witnessed several pioneering approaches leveraging image processing and machine learning techniques, including filtering [5], [6], [7], human eye visual modeling [8], [9], and low-rank matrix [10], [11], [12], [13], [14].

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However, these conventional methods exhibit notable limitations. Filter-based methods, such as top-hat filters [6] and max-median/max-mean filters [7], are only capable of suppressing simple background clutter; however, they often suffer from a sharp degradation in performance in complex scenarios, thereby leading to a high false alarm rate. While human-like visual system-based methods [8], [15], [16] are sensitive to small, high-brightness targets, yet when the target is similar to the background, it becomes difficult to distinguish. Low-rank matrix-based methods [10], [11], [12], [13] can adapt to images of dim and small targets, yet they still exhibit a high false alarm rate when detecting images containing small targets with varying shapes in complex backgrounds. Furthermore, most traditional approaches rely heavily on handcrafted features that underperform in challenging cases [17], [18], [19].

With the success of deep learning across diverse domains, new solutions have emerged for IRSTD [20]. Convolutional neural networks (CNNs) can effectively extract features of small targets from infrared imagery, thereby identifying true targets [21]. To improve small-target detection under complex backgrounds, many approaches adopt U-shaped architectures, yielding more robust detection performance [22]. Because small targets occupy few pixels, inconsistencies can arise between low-level details and high-level semantics. Accordingly, a number of methods introduce feature fusion modules to align shallow and deep representations [23], [24], [25]. In infrared images, the extremely small proportion of target pixels makes it difficult for networks to learn sufficiently informative features, prompting the design of new feature-extraction modules that enhance sensitivity to small targets [22], [25], [26], [27]. Other works propose revised network topologies to improve information flow and compensate for deficiencies in deep-layer features.

Despite these advances, the inherent locality of CNNs limits access to global context, motivating hybrid CNN–Transformer models that capture more discriminative features and thereby strengthen detection. Although existing CNN-based IRSTD methods have achieved strong results, they generally rely on fixed convolutional kernels for feature extraction, which unavoidably introduce background clutter and degrade performance. This problem is particularly acute when the target size is smaller than the kernel, at which point the convolution becomes dominated by background responses rather than the desired target, hindering the extraction of discriminative features. Adapting to small targets of diverse shapes to fully capture their weak infrared signatures is therefore crucial for accurate IR small-target detection.

Unlike generic objects (e.g., cars and faces) that possess rich geometric structures and textures, infrared small targets often appear as point-like blobs lacking clear shape information. Standard convolutions with fixed grid structures (e.g., 3×3) inevitably include a large proportion of background pixels when the target size is smaller than the receptive field (subkernel scale). This introduces overwhelming background noise that submerges the weak target signal. Therefore, our motivation for introducing Deformable Convolution is not to handle geometric transformations (like rotation or scaling), but to adaptively concentrate the sampling points onto the sparse target area. This acts as a spatial attention mechanism that physically filters out background clutter before the multiplication operation.

To visually demonstrate the effectiveness of our approach, we present a schematic comparison in Fig. 2. For subpixel infrared small targets, a standard fixed 3×3 convolution inevitably includes a majority of background pixels in its receptive field, introducing significant noise. In contrast, our proposed method generates centripetal offsets that force the sampling points to converge toward the target center. By physically excluding background pixels from the sampling set, our method ensures that the contribution of background clutter approaches zero. This mechanism allows enhanced dynamic deformable network (EDDNet) to significantly suppress, rather than merely filter, the background noise at the source.

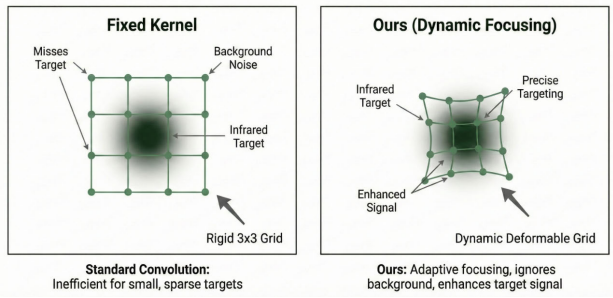


Fig. 2. Motivation comparison. (Left) Fixed 3×3 kernels inevitably introduce background noise on small targets. (Right) Our method concentrates sampling points toward the target center, effectively "shrinking" the kernel to exclude interference.

In this article, we propose an EDDNet to address IRSTD challenges. We develop a UNet-based architecture integrating three key components: 1) a dynamic deformable convolution module that selectively covers target regions while suppressing background noise during feature extraction; 2) a direction-aware offset enhancement (DAOE) module, strengthening sampling point deviation toward targets through boundary localization; and 3) a semantic refinement (SR) module bridging encoder–decoder transitions to enhance deep feature semantics. In summary, our contributions are as follows.

1) We propose EDDNet, a new CNN that can reduce the introduction of background noise during feature extraction. EDDNet overcomes the introduction of background noise caused by fixed convolution kernels in the feature extraction process of traditional convolution and further adapts to small targets of different sizes by introducing input-dependent dynamic convolution kernel sizes.
2) We propose a direction-aware offset enhancement module to enhance the offset capability of convolution sampling points in deformable convolutions. This module enhances the ability of convolutional kernel sampling points to bias toward small targets by locating the boundaries of small targets through one-way pooling.
3) We proposed an SR module during the transition from the encoding stage to the decoding stage, which refocuses the semantic information in the deep network model on small targets.

II. Related Work

During infrared imaging processes, the signal intensity received by infrared detectors for long-range targets is significantly attenuated due to atmospheric scattering and refraction, lens contamination, optical defocusing, and lens deformation [28], [29]. Infrared small targets are formally defined by SPIE [30] (International Society for Optics and Photonics) as objects occupying no more than 9×9 pixels. Three

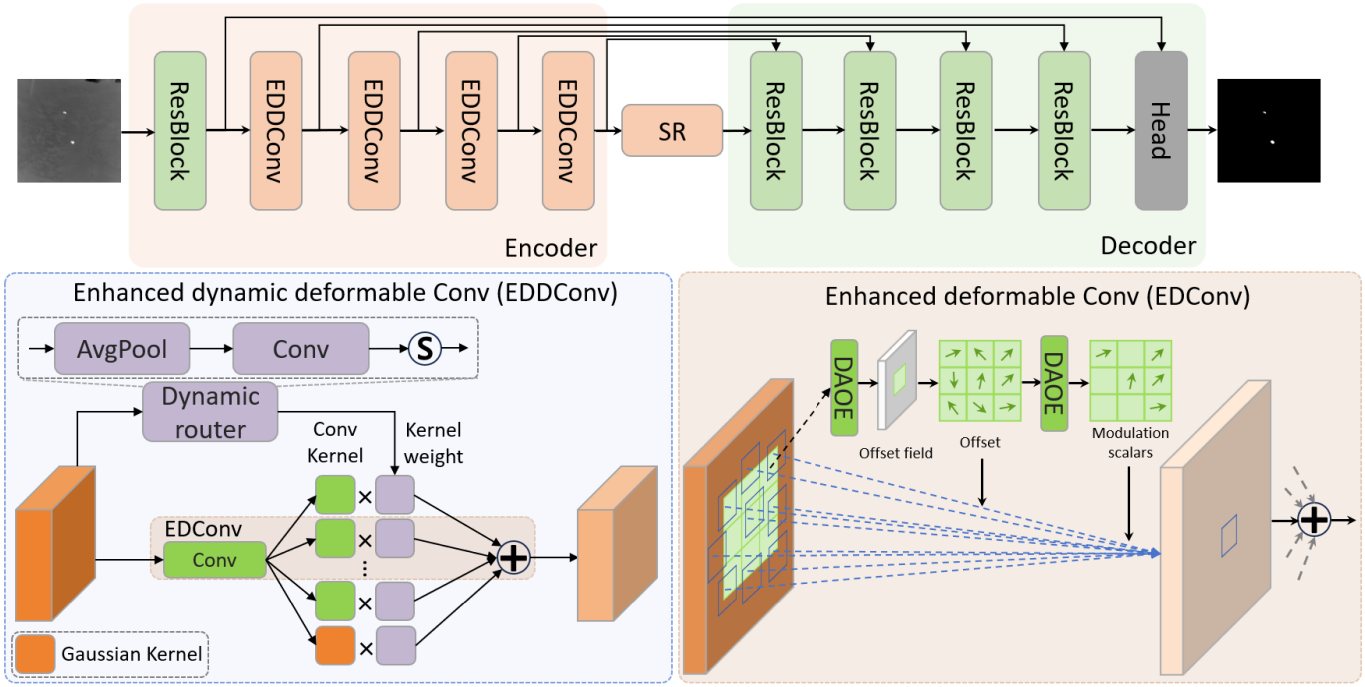


Fig. 3. Proposed EDDNet pipeline is illustrated. The UNet architecture is employed as the backbone network, where enhanced deformable dynamic convolution is introduced during the encoding stage to suppress background noise incorporation during feature extraction. Additionally, an SR module is incorporated at the encoder–decoder transition to maintain focus on both background semantics and small targets.

representative small-target images with varying scales are presented in Fig. 1. It can be observed that these infrared targets exhibit subconvolutional dimensions, generally smaller than standard 3×3 convolutional kernels. Furthermore, the combined effects of thermal diffusion and point spread phenomena result in extremely low saliency for small targets in infrared images, posing significant challenges for extracting adequate discriminative features.

With the remarkable success of deep-learning-based object detection methodologies, numerous frameworks originally designed for general vision tasks have been transferred to IRSTD [31], [32], [33], [34], [35], [36], [37]. However, these adapted frameworks often encounter challenges when confronting the extreme scale of small targets. To address these limitations, a wave of specialized networks tailored for IRSTD has emerged. These approaches typically formulate small-target detection as a binary segmentation problem [38], primarily relying on the U-Net architecture [21] and its variants [39], [40]. Dai et al. [23], [24] enhanced target information extraction through asymmetric designs and attention mechanisms focusing on target contrast features. Zhang et al. [41], [42], [43] initiated their methodology by mathematically characterizing small targets based on their morphological properties, subsequently exploring sparse target localization and feature refinement from a super-resolution perspective. Li et al. [17] developed a densely nested structure to preserve target responses. Zhang et al. [44] addressed target loss during downsampling through hierarchical feature fusion. Wu et al. [22] introduced a multilevel UIU-Net framework with auxiliary supervision. These efforts aim to amplify target characteristics and mitigate response degradation through network depth. Recently, the field of IRSTD has witnessed the emergence of several innovative directions. For instance, DSAD [45] introduces a dual-stream attention decoupling framework to better separate target features from background noise. Similarly, SPMix-Q [46] explores superpixel-based mixing strategies to enhance data augmentation and feature robustness. These emerging approaches indicate a trend toward more refined feature decoupling and data-efficient learning strategies, which further motivates our exploration of dynamic deformation for precise feature alignment in this work. Nevertheless, conventional convolutions with fixed 3×3 kernels remain universally employed, wherein background noise is inevitably introduced during feature extraction due to the intrinsic size disparity between convolutional operators and subkernel-scale targets.

III. Methodology

A. Network Architecture

Fig. 3 illustrates the overall architecture of our EDDNet, which primarily consists of an encoder–decoder structure with a segmentation head. The design is motivated by the observation that infrared small targets typically occupy only a few pixels in images, where conventional convolutions tend to overshoot them due to kernel sizes exceeding target dimensions. This spatial mismatch results in background coverage by redundant kernel regions, inevitably introducing background noise that hinders sufficient feature extraction for small targets. To address this, traditional convolutions are replaced with deformable convolutions to reduce background noise through adaptive sampling point offsets. However, existing deformable convolutions exhibit limited capability in capturing infrared small-target-oriented offsets through their sampling points. A DAOE module is therefore proposed to bias sampling points toward small targets, while modulation factors are introduced for each sampling point to mitigate adverse effects caused by erroneous offsets. Additionally, an SR module is implemented at encoder–decoder transitions, which employs dilated depthwise convolutions with varying rates and channel shuffling

operations to refine semantic information while enhancing model generalization [28].

B. Enhanced Deformable Dynamic Convolution

Given an infrared image $X \in \mathbb{R}^{3 \times H \times W}$, the channels are first expanded to C (where C is set to 16) using ResNet blocks. Enhanced deformable dynamic convolution is then employed to extract hierarchical features through downsampling. Specifically, the feature F_X is initially fed into a dynamic routing mechanism that generates path weights W_d. This dynamic routing process consists of max-pooling, convolutional operations, and sigmoid activation

$$W_d = Sigmoid\left(Conv2d\left(MaxPool\left(F_x\right)\right)\right). \tag{1}$$

Here, $MaxPool$ denotes max-pooling, $Conv2d$ represents 2-D convolution, and $Sigmoid$ indicates the sigmoid function. Local features under different receptive fields can be captured through multiscale convolutions to suppress background noise, which is critically important for infrared small targets immersed in substantial background interference. The kernel sizes of multiscale convolutions are configured as $\{1, 3, 5\}$. Since small targets typically occupy approximately 50 pixels on average, the use of larger kernels is avoided to prevent inevitable background noise introduction and unnecessary computational latency. Additionally, a learnable Gaussian convolution kernel is incorporated due to its demonstrated effectiveness in enhancing point-like infrared small targets. The convolution process is implemented using enhanced deformable convolution to reduce background noise contamination, formally expressed as follows:

$$\begin{aligned} F_{x1}, F_{x3}, F_{x5} &= EDConv2d_k\left(F_x\right), \quad k \in 1, 3, 5 \\ F_g &= Conv2d_{gas}\left(F_x\right) \\ F_d &= cat\left(F_{x1}, F_{x3}, F_{x5}, F_g\right) * W_d. \end{aligned} \tag{2}$$

Here, F_{x1}, F_{x3}, and F_{x5} denote features corresponding to different convolution kernels processed through enhanced deformable convolution, while F_g represents features obtained via standard convolution with a Gaussian kernel. $Conv2d_{gas}$ indicates learnable Gaussian convolution. $EDConv2d_k$ denotes the enhanced deformable convolution, which will be detailed in Section III.

C. DAOE Module

Since infrared small targets are extremely tiny, the original deformable convolution struggles to perceive their shapes and positions, potentially leading to offset failure and adverse effects on feature extraction. To address this, we propose a DAOE module, which is specifically designed to constrain the degrees of freedom of the learned offsets, encouraging sampling points to converge toward high-response centers (the targets) rather than diverge, thereby ensuring that the deformable convolution kernel spatially "shrinks" to better fit extremely small targets.

For standard convolution, a convolution operation can be decomposed into two steps: 1) sampling features on the feature map according to the kernel shape and 2) performing a Hadamard product between the sampled points and the feature map. In deformable convolution, our sampling points are not kernel-shaped but arbitrary. Taking a 3×3 convolution as an example, our sampling points are

$$S = \{(-1, -1), (-1, 0), \ldots, (0, 1), (1, 1)\}. \tag{3}$$

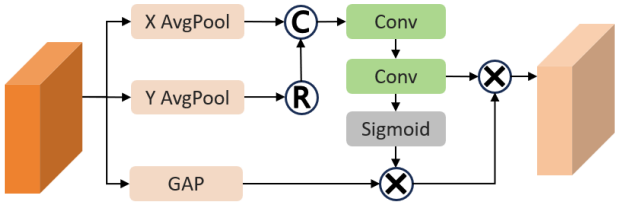


Fig. 4. Block diagram of the DAOE module.

For each sampled point value s_0, we have an initial position S_i and corresponding offset positions $\{\Delta b_n | n = 1, \ldots, N\}$, where $N = |S|$. The offset positions are implemented through our DAOE module. The convolution process can be formulated as follows:

$$y\left(b_0\right) = \sum_{b_n \in S} M\left(b_n\right) \cdot w\left(b_n\right) \cdot x\left(b_0 + b_n + \Delta b_n\right). \tag{4}$$

Additionally, to mitigate the negative impact of incorrect offsets on convolution results, we introduce a modulation factor $M(b_n)$ for each sampling point, which enhances or suppresses the influence of individual sampling points on the final convolution output.

The DAOE module employs directional max-pooling across different orientations to enhance small-target perception, as shown in Fig. 4. The specific expression is

$$\begin{aligned} F_{gap} &= MaxPool\left(F_x\right) \\ F_{gapxy} &= cat\left(MaxPool_x\left(F_x\right), reshape\left(MaxPool_y\left(F_x\right)\right)\right) \\ F_{convxy} &= Conv2d\left(F_{gapxy}\right) \\ F_{out} &= F_{convxy} * Sigmoid\left(Conv2d\left(F_{convxy}\right)\right). \end{aligned} \tag{5}$$

Here, $MaxPool_x$, $MaxPool_y$, and $MaxPool$ denote global max-pooling along the x-direction, the y-direction, and global max-pooling, respectively. F_{gap}, F_{gapxy}, F_{convxy}, and F_{out} are defined as the globally averaged pooling features, concatenated features from directional max-pooling, convolutional features, and final fused features, respectively. The mathematical validity of this design lies in the orthogonal decomposition of target localization. For infrared small targets, which often lack texture but possess high local intensity, standard global pooling discards spatial coordinates. In contrast, $MaxPool_x$ and $MaxPool_y$ preserve the peak response locations along the horizontal and vertical axes, respectively. The concatenation of these directional features acts as a spatial coordinate system, providing an explicit geometric prior. This guides the learnable offsets Δb_n to converge toward the intersection of the x and y peaks (i.e., the target center), thereby physically implementing the "focusing" effect illustrated in Fig. 2.

D. SR Module

Following feature extraction through the encoder stage, composed of enhanced dynamic deformable convolution, most background noise has been effectively filtered. As target features propagate to deeper network layers, their significance becomes increasingly prominent. At this stage, where feature map resolution is substantially reduced, an SR module is designed for the precise processing of these high-level features. As illustrated in Fig. 5, the SR integrates channel shuffling operations with depth-wise convolutions of varying receptive fields. Given that the majority background interference has been eliminated during encoding, the expanded

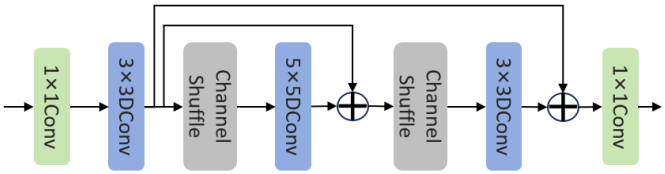


Fig. 5. Flowchart of the SR module.

receptive fields prove advantageous for suppressing residual noise surrounding small targets without introducing new background artifacts. Channel shuffling facilitates feature transition between the encoder and decoder stages while enhancing the model's representational capacity.

IV. Experiments

A. Datasets

To thoroughly validate our proposed method, this study employs three widely used datasets: NUAA-SIRST (427) [23], NUDT-SIRST (1327) [17], and IRSTD-1K (1000) [41]. These datasets encompass a variety of complex background scenarios, including urban scenes, countryside fields, bright regions, marine scenes, clouds, and background clutter, along with different target quantities and scales. To ensure fair comparison and reproducibility, the original dataset partitioning schemes are adopted, using split ratios of 50:50, 50:50, and 80:20 for training and validation, respectively. The NUAA-SIRST and NUDT-SIRST are resized to 256×256, while IRSTD-1K is resized to 512×512 for comprehensive evaluation of the proposed method.

B. Implementation Details

Experimental evaluations are conducted on a Windows 11 system fit with an Intel i9-12900K CPU, 32 GB RAM, and a GeForce RTX 3090 GPU. The model is implemented using PyTorch, with weights and biases initialized using Xavier initialization. During training, the Adagrad optimizer is used in combination with the SoftIoU loss function for parameter optimization with a learning rate of 0.05. Our model is trained from scratch.

C. Evaluation Metrics

CNN-based segmentation methods are primarily evaluated using pixel-level metrics, such as intersection over union (IoU) and normalized IoU (nIoU), alongside object-level metrics including the detection rate (P_d) and false alarm rate (F_a). These metrics collectively reflect the model's performance in both localization precision and target recognition.

IoU: This metric evaluates the algorithm's ability to accurately delineate target boundaries by calculating the ratio of the intersection area to the union area between the predicted region and the ground truth

$$\text{IoU} = \frac{\sum_{j=1}^{M} \text{TP}^{\text{j}}}{\sum_{j=1}^{M} \left(\text{TP}^{j} + \text{FP}^{j} + \text{FN}^{j}\right)} \tag{6}$$

where M denotes the number of samples and TP, FP, and FN represent true positives, false positives, and false negatives at the pixel level, respectively.

nIoU: Specifically developed for infrared small-target segmentation, nIoU serves as a normalized variant of the standard IoU. This metric is designed to mitigate the influence of target size variations on the evaluation weighting, thereby providing a more balanced assessment of segmentation performance for targets of different scales

$$\text{nIoU} = \frac{1}{M} \sum_{j=1}^{M} \frac{\text{TP}^{j}}{\text{TP}^{j} + \text{FP}^{j} + \text{FN}^{j}}. \tag{7}$$

Detection Rate (P_d): This metric evaluates the algorithm's ability to correctly identify targets. It is defined as the ratio of true positives (TPs) to the actual target number [sum of TP and false negatives (FNs)] across all samples

$$P_d = \frac{\sum_{i=1}^{N} \text{TP}^{i}}{\sum_{i=1}^{N} \left(\text{TP}^{i} + \text{FN}^{i}\right)}. \tag{8}$$

Here, the numerator represents the count of correctly detected targets across all N samples, while the denominator represents the ground-truth total. A higher P_d indicates superior sensitivity in target capture.

False Alarm Rate (F_a): This metric quantifies the frequency of incorrect target assignments. It is defined as the ratio of false positives (FPs) to the total number of predicted targets (sum of TP and FP) across all samples:

$$F_a = \frac{\sum_{i=1}^{N} \text{FP}^{i}}{\sum_{i=1}^{N} \left(\text{TP}^{i} + \text{FP}^{i}\right)}. \tag{9}$$

Here, the numerator represents the number of falsely detected targets across all N samples, while the denominator denotes the total number of predicted targets. A lower F_a value signifies a robust capability to suppress background clutter and sensor noise.

D. Quantitative Results

For every algorithm, its results were replicated by means of an integrated toolbox, where parameter configurations were kept consistent with the descriptions provided in their individual articles. Quantitative results are tabulated in Table I. To comprehensively validate the superiority of EDDNet, we compared it against ten state-of-the-art (SOTA) methods, including classic algorithms and recently published approaches such as RRCANet [48], FMNet [49], and SCTransNet [50].

As shown in Table I, on the NUAA-SIRST dataset, our method achieves an IoU of 79.289% and an nIoU of 76.910%, exceeding the second-best method (MRF-Net) by 1.37% and 1.32%, respectively. Although P_d (96.591%) is 0.36% lower than the highest value reported by FMNet, EDDNet maintains a comparable F_a of 15.897, indicating a balanced performance across detection accuracy and pixel-level segmentation quality. On the NUDT-SIRST dataset, EDDNet attains an IoU of 94.762% and an nIoU of 91.919%, representing improvements of 0.54% and 0.52% over MRF-Net. P_d reaches 99.361%, which is 0.1% lower than DNA-Net, while F_a remains at 3.913, reflecting stable detection performance with controlled false alarms. In the IRSTD-1K dataset, EDDNet achieves an IoU of 68.564%, which is 0.125% higher than PBT. Although its nIoU (66.507%) is slightly lower than that of PBT (67.841%), P_d improves to 94.913% and F_a decreases to 16.154, both of which outperform PBT. Overall, EDDNet demonstrates a more balanced performance in terms of object overlap, detection rate, and false alarm control. Finally, on the WideIRSTD-Full dataset, EDDNet achieves an *IoU* of 46.533% and an *nIoU* of 33.426%, exceeding DNA-Net and

TABLE I

IoU (%), nIoU (%), PD (%), AND FA ($\times 10^{-6}$) FOR VARIOUS SOTA METHODS OVER THE NUAA-SIRST, NUDT-SIRST, IRSTD-1K, AND WIDEIRSTD-FULL DATASETS

Methods	Params	FLOPs	NUAA-SIRST (Tr=50%)				NUDT-SIRST (Tr=50%)				IRSTD-1K (Tr=80%)				WideIRSTD-Full (Tr≈ 82%)			
			IoU	nIoU	Pd	Fa	IoU	nIoU	Pd	Fa	IoU	nIoU	Pd	Fa	IoU	nIoU	Pd	Fa
ACM [23]	0.398M	0.402G	69.44	67.357	92.015	22.707	64.855	62.909	96.72	28.587	60.326	58.516	93.266	68.494	35.855	23.544	39.571	42.236
ALCNet [24]	0.427M	**0.378G**	61.047	59.216	87.072	55.978	61.131	59.297	97.249	29.093	58.088	56.345	92.929	74.453	36.994	25.625	45.221	54.161
DNA-Net [17]	4.697M	14.261G	74.815	72.571	93.536	38.279	94.192	91.366	99.259	**2.436**	65.735	63.763	89.562	**12.336**	45.991	32.834	56.487	27.084
ISNet [41]	0.966M	30.618G	70.491	68.376	95.057	67.983	81.236	78.799	97.778	6.343	61.852	59.996	90.236	31.561	40.430	28.919	47.980	38.581
UIU-Net [22]	50.540M	54.426G	77.531	75.205	92.395	**9.33**	90.517	87.801	98.836	8.342	65.69	63.719	91.246	13.475	37.420	26.564	48.068	**20.088**
PBT [47]	20.411M	31.831G	76.737	74.435	96.057	25.158	91.419	88.676	98.336	14.845	68.439	67.841	94.205	18.307	41.046	29.701	52.968	26.885
MRF^{3}Net [25]	0.538M	8.295G	77.928	75.590	96.198	29.897	94.221	91.394	98.919	3.769	67.014	65.004	94.657	20.437	44.516	31.850	56.234	28.427
RRCANet [48]	**0.202M**	1.04G	72.404	70.229	95.437	16.328	84.587	84.860	97.671	5.032	67.485	61.158	94.557	16.473	39.748	29.016	54.513	22.033
FMNet [49]	3.606M	14.171G	75.052	72.667	96.958	11.194	80.510	82.041	97.671	9.536	63.632	61.258	90.816	58.454	41.050	30.728	55.661	35.552
SCTransNet [50]	11.2M	40.43G	75.478	72.932	95.174	9.581	84.886	85.440	97.777	6.779	64.807	62.355	91.836	14.195	42.967	32.146	**59.435**	33.592
Ours	3.794M	15.228G	**79.289**	**76.910**	**96.591**	15.897	**94.762**	**91.919**	**99.361**	3.913	**68.564**	**66.507**	**94.913**	16.154	**46.533**	**33.426**	57.360	59.883

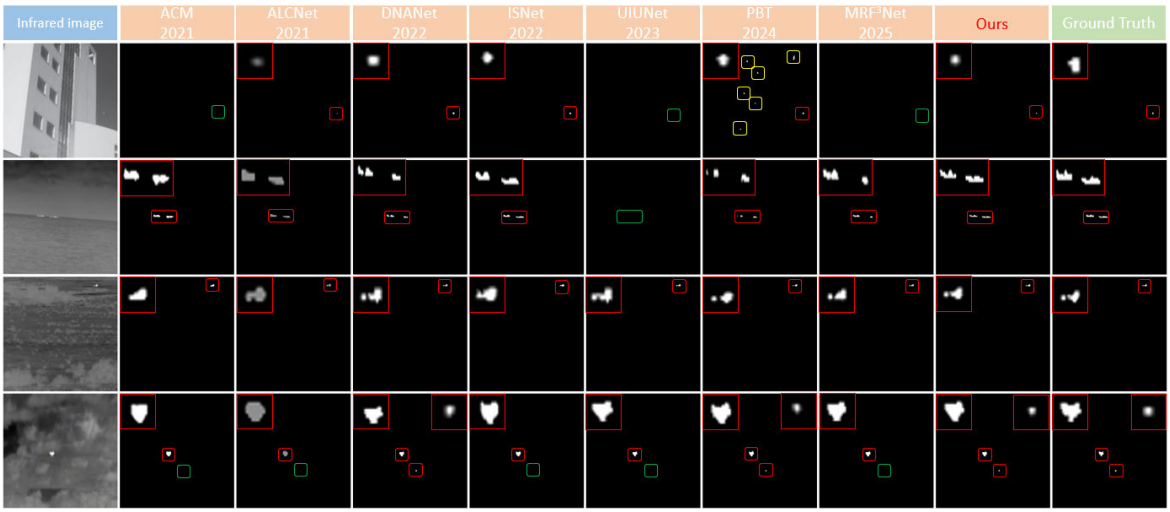


Fig. 6. Qualitative results of different methods on the NUAA-SIRST dataset. Red, green, and yellow boxes highlight correctly detected targets, missed detections, and false alarms, respectively.

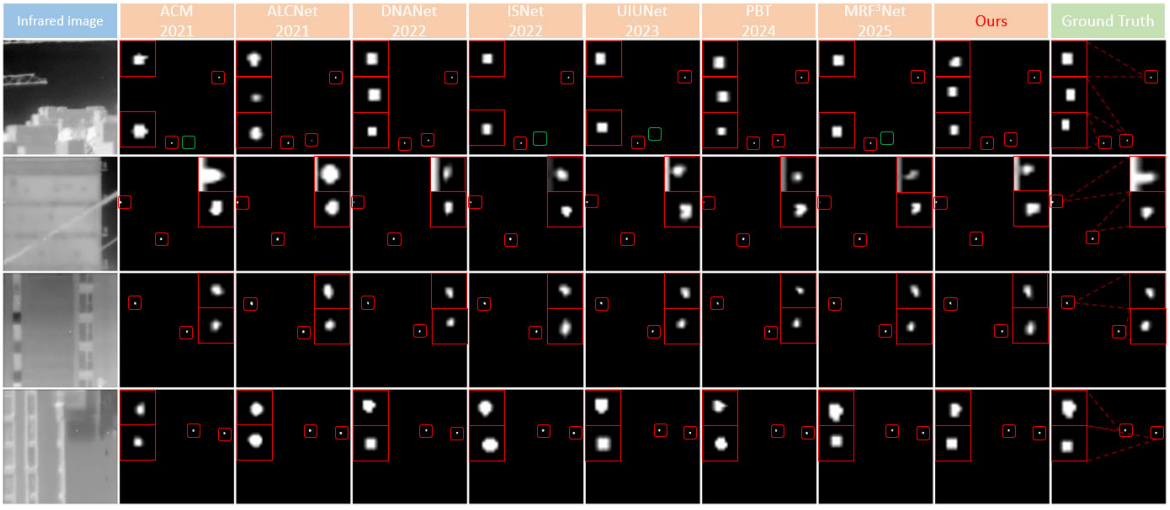


Fig. 7. Qualitative results of different methods on the NUDT-SIRST dataset. Red, green, and yellow boxes highlight correctly detected targets, missed detections, and false alarms, respectively.

SCTransNet by 0.54% and 0.59%, respectively. P_d remains at 57.360%, while F_a is maintained at 59.883 $\times$ 10^{-6}, indicating consistent performance under large-scale and complex conditions.

E. Qualitative Results

Qualitative outcomes for the NUAA-SIRST dataset are presented in Figs. 6–8, while the corresponding 3-D visualization results are displayed in Figs. 9–11. Against SOTA infrared small-target segmentation algorithms, our method yields higher-precision target localization and shape delineation while maintaining acceptable false alarm rates. This enhancement is attributable to our innovative design of the convolutional process, which suppresses the introduction of background noise and thus enables effective handling of infrared small targets of varying sizes in diverse cluttered

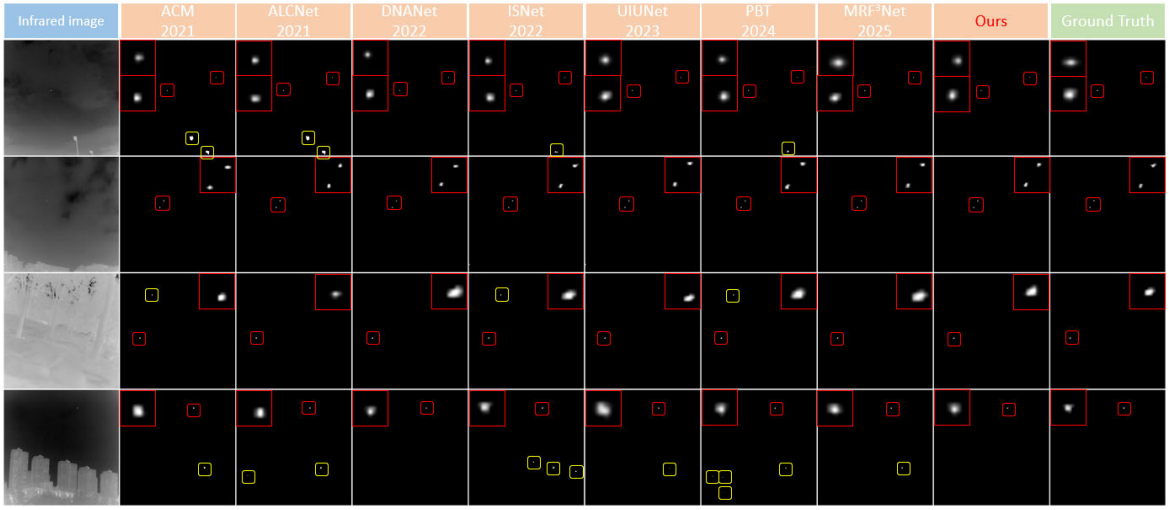


Fig. 8. Qualitative results of different methods on the IRSTD-1K dataset. Red, green, and yellow boxes highlight correctly detected targets, missed detections, and false alarms, respectively.

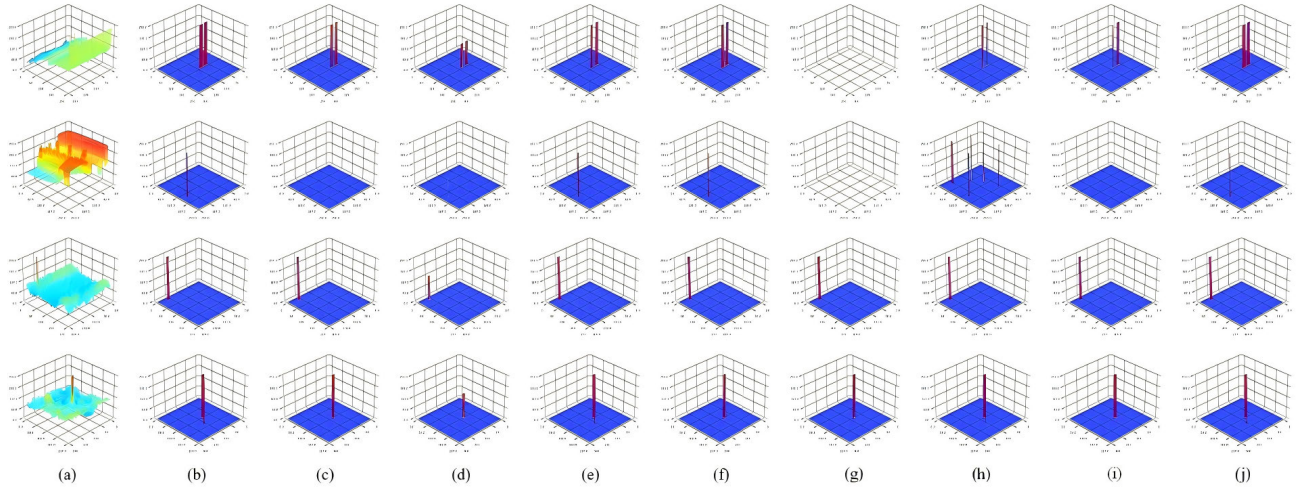


Fig. 9. 3-D visualization of different methods on the NUAA-SIRST dataset. (a) Original infrared image. (b) Ground truth. (c) ACM. (d) ALCNet. (e) DNANet. (f) ISNet. (g) UIU-Net. (h) PBT. (i) MRF3Net. (j) Ours.

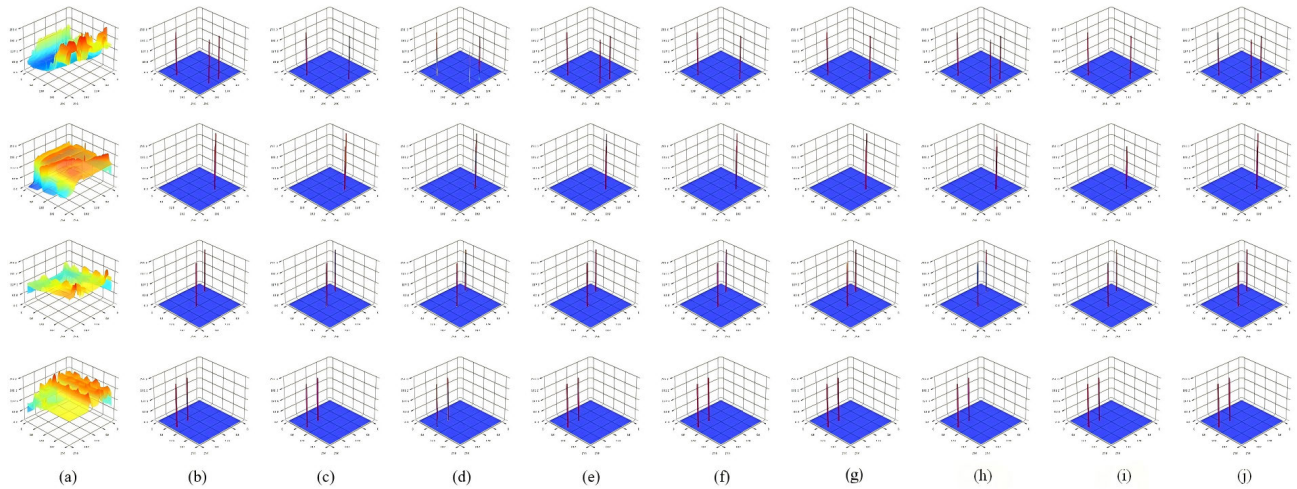


Fig. 10. 3-D visualization of different methods on the NUDT-SIRST dataset. (a) Original infrared image. (b) Ground truth. (c) ACM. (d) ALCNet. (e) DNANet. (f) ISNet. (g) UIU-Net. (h) PBT. (i) MRF3Net. (j) Ours.

backgrounds and under noise interference. For infrared images containing small targets, heatmaps depicting feature activation patterns across various layers of the EDDNet model can be found in Fig. 12. The heatmaps demonstrate that our EDDNet

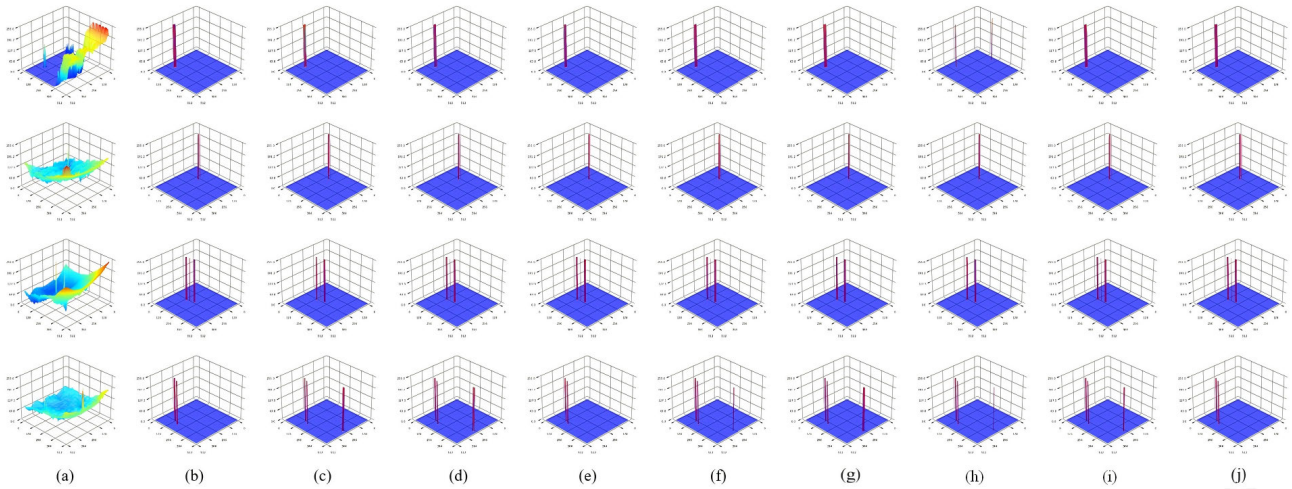


Fig. 11. 3-D visualization of different methods on the IRSTD-1K dataset. (a) Original infrared image. (b) Ground truth. (c) ACM. (d) ALCNet. (e) DNANet. (f) ISNet. (g) UIU-Net. (h) PBT. (i) MRF3Net. (j) Ours.

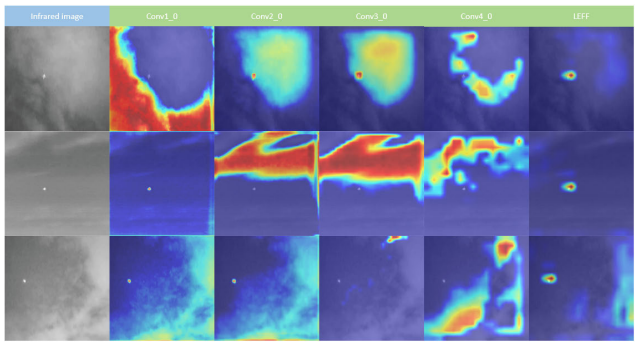


Fig. 12. Several visualization examples from the NUAA-SIRST dataset. Evidently, the deep network before and after SR refocuses on small targets.

TABLE II

IoU (%), nIoU (%), PD (%), AND FA ($\times 10^{-6}$) VALUES OBTAINED THROUGH THE MAIN VARIANTS OF EDDNET ON THE NUAA-SIRST DATASET

Methods	Params	FLOPs	NUAA-SIRST			
			IoU	nIoU	Pd	Fa
baseline (resnet18)	4.086M	**7.041G**	75.910	73.961	95.198	22.707
+ dynamic router	15.444M	25.381G	77.385	75.063	95.991	17.835
+ deformable conv	**2.491M**	14.884G	76.984	74.674	95.674	18.894
+ DAOE	3.221M	15.082G	78.917	76.549	96.417	17.045
Ours (+ SR)	3.794M	15.228G	**79.289**	**76.910**	**96.591**	**15.897**

model exhibits superior activation concentration around small targets in most layers, indicating effective focus on the critical regions. Additionally, the SR module demonstrates excellent semantic discrimination capability, where deeper network layers progressively suppress background interference while maintaining precise attention to target neighborhoods.

F. Ablation Study

To verify the effectiveness of each component in EDDNet, we conducted a comprehensive ablation study on the NUAA-SIRST dataset. The quantitative results are summarized in Table II.

1) Impact of Dynamic Routing: As shown in the second row of Table II, the integration of the dynamic router with multiscale convolutions yields an IoU of 77.38% and an nIoU of 75.06%, outperforming the baseline (ResNet18) by 1.47% and 1.15%, respectively. This quantitative improvement indicates that dynamically adjusting weights for different kernel sizes based on input features effectively enhances target representation.

2) Impact of Deformable Convolution: We further evaluated the impact of replacing standard convolutions with deformable convolutions. As observed in Table II (+deformable conv), naively applying deformable convolution results in a slight performance drop, with IoU decreasing to 76.98%. This suggests that without proper constraints, the learned offsets may become disordered on tiny subpixel targets, failing to extract effective features.

3) Impact of the DAOE Module: To address the offset issue, we introduced the DAOE module. The quantitative comparison in Table II (+DAOE) demonstrates a significant recovery and boost in performance: the IoU increases by 1.93% (from 76.98% to 78.91%) and nIoU improves by 1.87%. This substantial gain proves that the proposed directional constraints effectively guide the sampling points to focus on the small targets.

4) Impact of the SR Module: Finally, the inclusion of the SR module (ours) achieves the best performance, reaching an IoU of 79.29% and the lowest false alarm rate of 15.897 ($\times 10^{-6}$). As visualized in the feature heatmaps, the SR module effectively suppresses background noise in deep layers, further refining the detection results.

V. CONCLUSION

This research presents an EDDNet for accurate infrared small-target segmentation. Built upon the UNet architecture, EDDNet adopts enhanced deformable dynamic convolutions to fully extract target features while reducing background noise contamination. The core innovation stems from the critical observation that infrared small targets typically occupy subkernel dimensions (e.g., smaller than 3×3 convolutional kernels), where conventional convolutions inevitably introduce background noise during feature extraction. Enhanced deformable convolution is therefore introduced to adaptively concentrate sampling points on target regions, enabling comprehensive extraction of small-target features. Furthermore,

a DAOE module is designed to strengthen the sampling points' positional bias toward target pixels. Finally, an SR module is integrated at the transitions between the encoder and the decoder. This module not only refines deep-layer semantic features but also enhances the model's robustness. Collectively, these enhancements enable the balanced, efficient extraction and integration of multilevel features for infrared small targets. Comprehensive evaluations on three widely used public benchmark datasets confirm EDDNet's superior performance. Our approach surpasses current SOTA methods, demonstrating exceptional efficacy in tackling various infrared small-target segmentation challenges involving targets of different types and scales. When compared with traditional convolution-based solutions, our approach demonstrates notable advantages, underscoring its potential for extensive applications in this field.

Despite the superior detection performance, our method has certain limitations. First, the introduction of the dynamic offset calculation mechanism inevitably increases the computational complexity compared to standard static convolutions, resulting in a slight decrease in inference speed. Second, although the proposed method significantly suppresses background noise, in extremely complex scenarios with high-intensity clutter that mimics target characteristics, a small number of false alarms may still occur. In future work, we will focus on model lightweighting and further exploring semantic constraints to address these challenges.

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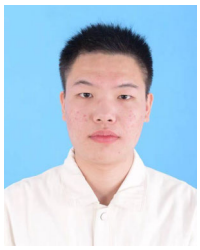
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